

REPORT IRL EXPERIMENTS

Impedance Spectroscopy



Skin Maven Bandage

ISEN MEDITERRANNEE
Master Project 2024-2025

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I. Introduction

During the first semester, we worked on the theoretical part of the project. Now, let's move on to the IRL part of our master project. The objective is to build our prototype to meet all the customers' requirements. But to achieve this, we had to cut our work into several parts.

First, we build our circuit in real life. To understand how everything works, before testing it on our skin, we tried with a RC circuit.

Then, we did some tests with 2 electrodes to have a first impression of the results we could get in real life.

After some tests with 2 electrodes, we decided to try with 4 electrodes because our project sponsor told us that with that, our measurements would be more precise.

With those 4 electrodes, we took several measures from 83 volunteers to build our database for our AI.

Then, we implemented our AI to train and then tested our data.

II. Material

- To build the RC circuit:
 - Resistor 1kOhm
 - Capacitor 1 nF
 - Soldering Iron
 - Tin
 - Wires
 - Box to contain the circuit
- To do the experiments:
 - HF2LI Lock-In Amplifier
 - HF2TA 50MHz Current Amplifier
 - Electrodes
 - Multimeter
 - Cables
 - Cables BNC-pince croco
 - Connectors male female
 - Computer: LabONE
- To build the database
 - Excel sheet: to store the biometrics of the volunteers (written at first in a google form)
 - CSV files: where the measured data was stored right after taking the IRL experiments
 - Excel sheet: to build our final database, containing some biometrics ad all the measured data
- To implement the AI
 - Computer: python
 - “Random Forest” algorithm

III. Tests with the RC circuit

a) Build the RC circuit

We first had to find the “Cut Frequency”.

To do so, we used the formula $F_c = 1 / (2\pi RC)$ to find the “cut frequency” with $F_c = 10\text{ K}$, and after that we continued our calculations to find the resistance.

$$10\,000 = 1 / (2\pi RC)$$

$$2\pi RC = 1 / 10\,000$$

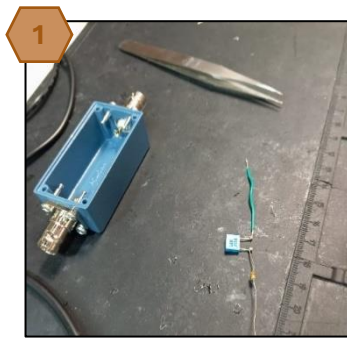
$$RC = 1 / 10\,000 \times 2\pi$$

$$RC = 1 / 62822 = 1,59 \times 10^{-5} \text{ with } C = 1 \times 10^{-9}$$

$$R = 1,59 \times 10^{-5} / C = 1,59 \times 10^{-5} / 1 \times 10^{-9} = 15\,900\text{ Ohm}$$

We'll take $R = 10\,000\text{ Ohm}$, which means that we are going to use a frequency F_c of **15 916 Hz**.

We assembled our RC circuit using a **1 kOhm** resistor and a **1 nF** capacitor, all in a close box that received the input from the oscilloscope and gives its output to the amplifier.



RC circuit sold with the box
where it will be closed in



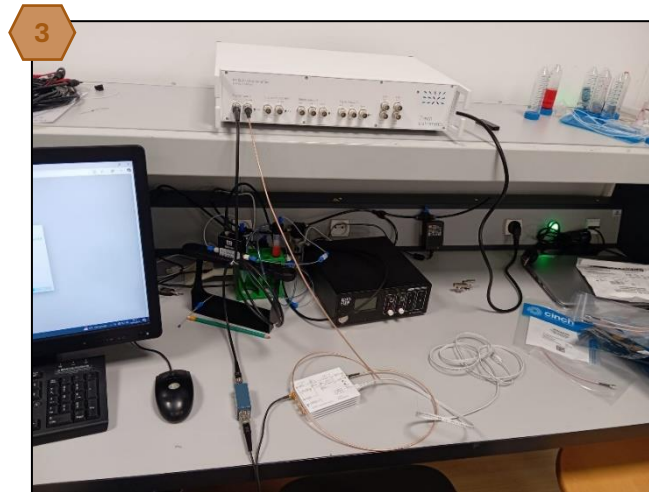
RC circuit sold in the box

After connecting the circuit and the machine to the PC and launching **LabOne**, we start the setup for the measurements. We start by setting-up the HFTA to control the amplifier and after that we go into the sweeper section and set the start and end frequency. We are going to use the following settings:

- Signal output: ON
- Amp = Ohm
- Signal input: AC
- Control:
 - o Start: 1 Hz
 - o Stop: 1 MHz
 - o Length: 100
 - o Sweep param: OSC 1 frequency
 - o Sweep mode: sequential
 - o Distribution: Logarithmic
 - o Demod: module (to have the phase)
 - o Amplitude: Checked
 - o Phase: Checked

After that we run the program by cleaning the history and doing a first single run.

b) Results with the RC circuit



Electronical circuit of our test with an RC circuit

After setting all the parameters on LabOne, we launched several simulations to see the tendency of the amplitude curve. We decided to start measuring the amplitude from 0 Hz to 1 MHz. But we observed that from 0 Hz to 100 Hz, the values were not interesting (near to 0 V). But from 100 Hz to 1 MHz, the amplitude values seem to follow the tendency of the amplitude that we wanted to get.

We will next try this experiment with 2 electrodes to see if we get the same amplitude tendency.

IV. Tests with 2 electrodes

a) Build the circuit with 2 electrodes

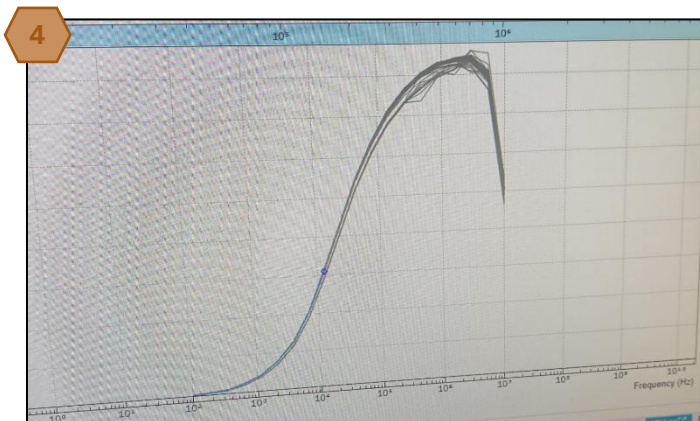
We started by detaching the circuit in the box and connecting the 2 cables equipped with clamps in its place. One of them was connected to the output of the oscillator and the others was connected to the amplifier through a T attachment.

On LabOne we have the following settings:

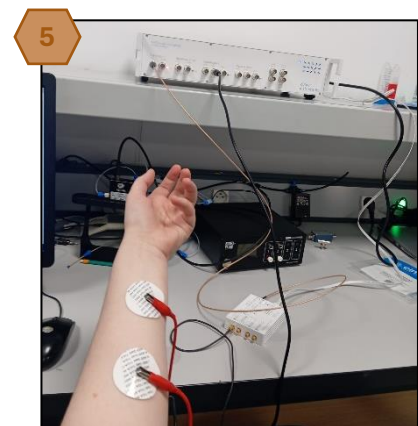
- Start: 100 Hz
- Stop: 1 MHz
- Length: 100
- Sweep param: OSC 1 frequency
- Sweep mode: sequential
- Distribution: Logarithmic (to see something)
- Demod: module (to have the phase)
- Amplitude: Checked
- Phase: Checked

We define “electrode of alimentation” the electrode connecting the “Signal Output 1” of the HF2LI Lock-In Amplifier to the skin, before any current comes through it. We define “electrode of measurement” the electrode that wires the skin to the HF2TA 50MHz Current Amplifier and the “Input 1” of the HF2LI Lock-In Amplifier. It measures the current once it goes through the skin.

b) Results obtained with the “2 electrodes circuit”



First amplitude curves obtained on LabOne with the electrodes attached to the skin of our patient



Electronical circuit of our test with electrodes attached to the skin

With the amplitude curves obtained on LabOne, we can observe that the values are becoming interesting at 100 Hz because we chose to start measuring the amplitude at 100 Hz (start frequency). We can suppose that the values are near 0 V from 0 Hz to 100 Hz by looking at the curve, the plateaus seems really plan at the beginning of our measurements (100 Hz).

Concerning the tendency of the curves, they all follow the same tendency. They seem right from 100 Hz to 10 000 000 Hz because the amplitude increases. But from 10 000 000 Hz to 1 MHz, the curve decreases. This is not supposed to happen for such a measurement of amplitude.

Maybe this issue is due to a misplacement of the electrodes. We will see if this happens again in our future experiments.

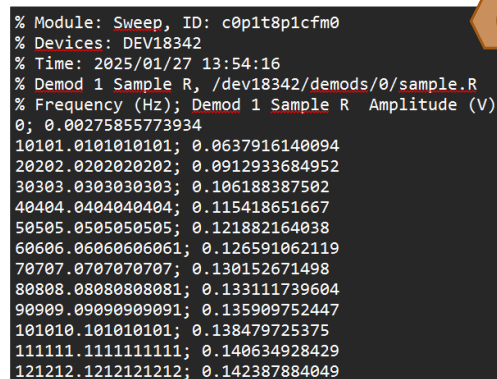
Now that we have done these experiments with 2 electrodes, we will have to find the resistance of contact. This means that we must find resistance between the electrodes and the skin.

c) Storage of the results of the “2 electrodes circuit”



At the beginning of this project, we did all our measurements with 4 electrodes. Later, we were told that it would be better to use 2 electrodes. We first made a python code for our measurements with 2 electrodes (cf. all the screenshots after). But after we changed our way of measuring our data (with 4 electrodes), we built a new code with another way of processing data. All the explanations in this part are based on the old version of the code. Only one screenshot (cf. 9') is related to the new version of the code.

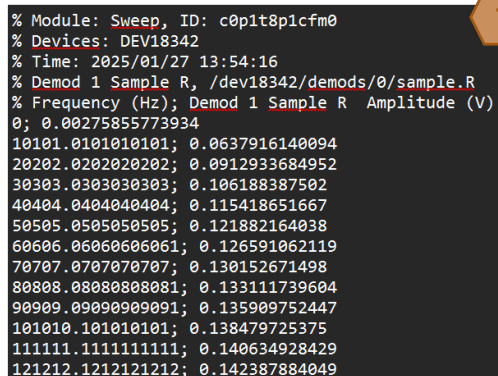
Once we are done measuring the amplitude and the frequency with the 2 electrodes attached to the skin, we get the values stored in a “.txt” file generated by LabOne. This file is our starting point to then be able to calculate the impedance of the skin, using the values that we measured on a patient.



```
% Module: Sweep, ID: c0p1t8p1cfm0
% Devices: DEV18342
% Time: 2025/01/27 13:54:16
% Demod 1 Sample R, /dev18342/demods/0/sample.R
% Frequency (Hz); Demod 1 Sample R Amplitude (V)
0; 0.00275855773934
10101.0101010101; 0.0637916140094
20202.0202020202; 0.0912933684952
30303.0303030303; 0.106188387502
40404.0404040404; 0.115418651667
50505.0505050505; 0.121882164038
60606.0606060606; 0.126591062119
70707.0707070707; 0.130152671498
80808.0808080808; 0.133111739604
90909.0909090909; 0.135909752447
101010.1010101010; 0.138479725375
111111.1111111111; 0.140634928429
121212.1212121212; 0.142387884049
```

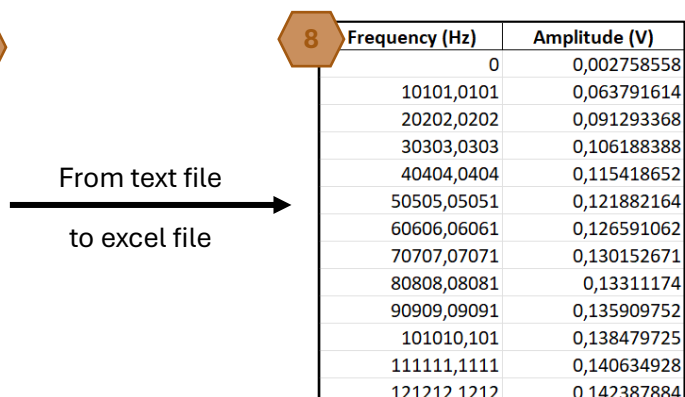
Example of obtained values
stored in a text file

Once we downloaded the text file, we take the values stored in it to store them in an excel file. This is being done thanks to a python code that we coded. Now, we have two different columns in our excel file; Frequency (Hz) and Amplitude (V).



```
% Module: Sweep, ID: c0p1t8p1cfm0
% Devices: DEV18342
% Time: 2025/01/27 13:54:16
% Demod 1 Sample R, /dev18342/demods/0/sample.R
% Frequency (Hz); Demod 1 Sample R Amplitude (V)
0; 0.00275855773934
10101.0101010101; 0.0637916140094
20202.0202020202; 0.0912933684952
30303.0303030303; 0.106188387502
40404.0404040404; 0.115418651667
50505.0505050505; 0.121882164038
60606.0606060606; 0.126591062119
70707.0707070707; 0.130152671498
80808.0808080808; 0.133111739604
90909.0909090909; 0.135909752447
101010.1010101010; 0.138479725375
111111.1111111111; 0.140634928429
121212.1212121212; 0.142387884049
```

Example of obtained values
stored in a **text file**



From text file
to excel file

Frequency (Hz)	Amplitude (V)
0	0,002758558
10101,0101	0,063791614
20202,0202	0,091293368
30303,0303	0,106188388
40404,0404	0,115418652
50505,0505	0,121882164
60606,0606	0,126591062
70707,0707	0,130152671
80808,0808	0,13311174
90909,0909	0,135909752
101010,101	0,138479725
111111,1111	0,140634928
121212,1212	0,142387884

Example of obtained values
stored in an **excel file**

After that, we will calculate the Intensity (A), the Impedance (Ohm), the Complex Impedance (Ohm) and the Phase (°). All of those will be calculated with our python code in which we will add few lines to obtain these new values.

```
data["Z_4E_XY"] = (data["Demod4 X (V)"] + 1j * data["Demod4 Y (V)"]) / ((data["Demod1 X (V)"] + 1j * data["Demod1 Y (V)"]) / Resistance)
data["Z_4E_XY_Mag"] = np.abs(data["Z_4E_XY"]) # Magnitude de l'impédance
```

```
# Add the "Intensity (A)" column (Amplitude divided by 100 Ohm)
df["Intensity (A)"] = df["Amplitude (V)"] / 100

# Add the "Impedance (Ohm)" column (Amplitude divided by Intensity)
df["Impedance (Ohm)"] = 1 / df["Intensity (A)"]

# Construct a complex impedance
reactance = df["Frequency (Hz)"] * 1e-6 # Assume a capacitance-like effect
df["Complex Impedance"] = df["Impedance (Ohm)"] + 1j * reactance # Create Z = R + jX

# Calculate Phase using np.angle()
df["Phase (degrees)"] = np.angle(df["Complex Impedance"], deg=True) # Extract phase in degrees
```

Code lines that calculate the **Intensity**, **Impedance**, **Complex Impedance** and **Phase** that will be stored in the same excel file

We decided to calculate Intensity (A) because we need it to calculate the Impedance (Ohm). And it is our main goal, we first calculate the Intensity. For that, we use this formula: **Intensity = Amplitude / 100**.

Once we calculate the Intensity (A), we can calculate the Impedance of the skin. To do that, we applied this formula: **Impedance = 1 / Intensity (A)**.

Now, we want to calculate the phase. But first, we must calculate the Complex Impedance because the phase is basically the imaginary impedance. This is why we use this formula to calculate the Complex Impedance:

$$Z = R + jX$$

$$\text{Complex Impedance} = \text{Impedance} + 1j * \text{reactance} \quad \text{with} \quad \text{reactance} = \text{Frequency} * 1e - 6$$

Finally, we can calculate the Phase (°) using this formula: Phase = Complex Impedance. We just have to translate the value in Ohm (of the complex impedance) in degrees (of the phase). For that, we used a python command in our code (cf 9).

Now, our excel file looks like this:

Frequency (Hz)	Amplitude (V)	Intensity (A)	Impedance (Ohm)	Complex Impedance	Phase (degrees)
0	0,002758558	2,75856E-05	36250,82723	(36250.82722536217	0
10101,0101	0,063791614	0,000637916	1567,604168	(1567.604167928162	0,000369191
20202,0202	0,091293368	0,000912934	1095,369813	(1095.369813254921	0,001056712
30303,0303	0,106188388	0,001061884	941,7225589	(941.7225588637605	0,001843681
40404,0404	0,115418652	0,001154187	866,4110918	(866.4110917576381	0,00267192
50505,05051	0,121882164	0,001218822	820,4645921	(820.4645920860278	0,003526936
60606,06061	0,126591062	0,001265911	789,9451851	(789.945185118966+	0,004395839
70707,07071	0,130152671	0,001301527	768,3284473	(768.3284472692261	0,005272767
80808,08081	0,13311174	0,001331117	751,2485397	(751.2485397418321	0,006163023
90909,09091	0,135909752	0,001359098	735,7823718	(735.7823717543483	0,007079141
101010,101	0,138479725	0,001384797	722,1273708	(722.1273708422098	0,008014448
111111,1111	0,140634928	0,001406349	711,0609087	(711.0609086738032	0,008953098
121212,1212	0,142387884	0,001423879	702,306946	(702.3069460431548	0,009888757

4 new columns in our excel file with calculated values

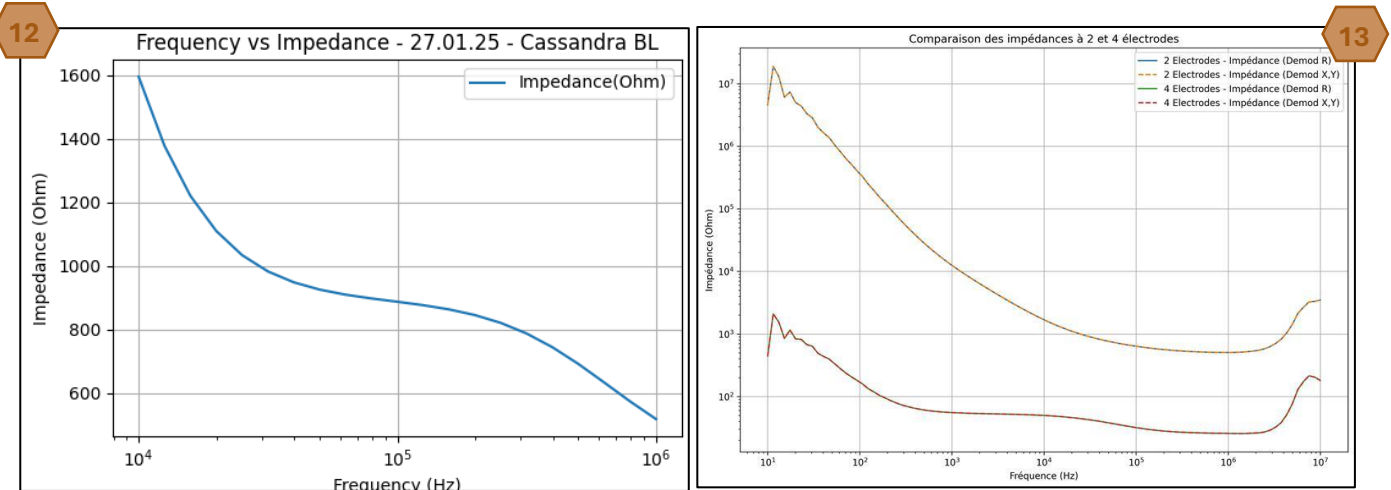
We now have 6 different columns, but we are interested in only 3 of them: Frequency (Hz), Impedance (Ohm) and Phase (°). To keep these 3 classes, we will copy paste the actual excel file and modify the new version by hand. We will delete the columns that are not interesting for us; Amplitude (V), Intensity (A), Complex Impedance (Ohm). Also, we will classify our values in lines and not in columns so it will be easier to train our AI. After doing that, this is the tab that we get:

11	Frequency (Hz)	0	10101,01	20202,02	30303,03	40404,04	50505,05	60606,06	70707,07	80808,08	90909,09	101010,1	111111,1	121212,1
	Impedance (Ohm)	36250,83	1567,604	1095,37	941,7226	866,4111	820,4646	789,9452	768,3284	751,2485	735,7824	722,1274	711,0609	702,3069
	Phase (degrees)	0	0,000369	0,001057	0,001844	0,002672	0,003527	0,004396	0,005273	0,006163	0,007079	0,008014	0,008953	0,009889

Final excel file with values that will be used to train our AI

V. Results with 2 electrodes

a) Impedance graph

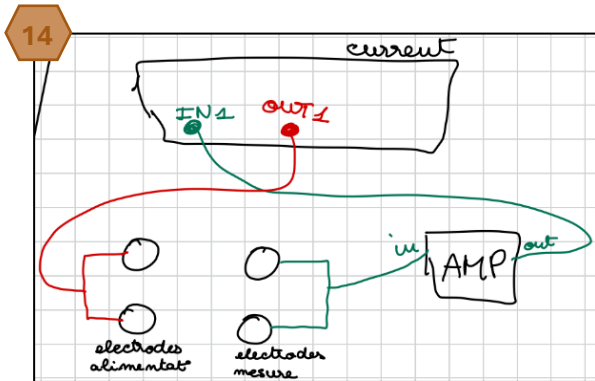


12. This is the graph that we obtained with our python code once all the data stored in an excel file were processed. We observe a great tendency for this curve. We visualize well the 2 plateaus that we obtained in our theoretical measurements. That means that all the work we did before was accurate compared to the reality.

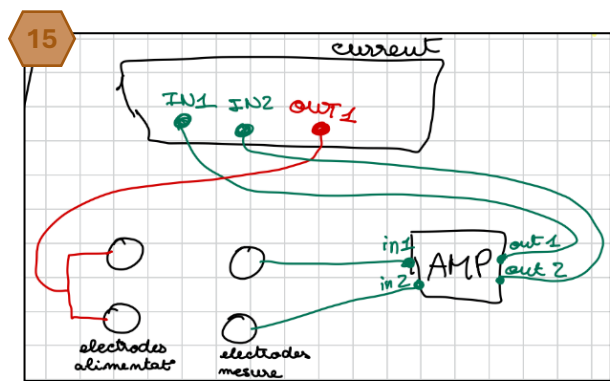
13. We get the same tendency with our new circuit and our new python code (*blue & yellow curve*). The 2 plateaus are less visible but this criteria varies with each person.

VI. Tests with 4 electrodes

a) Theoretical circuit with 4 electrodes



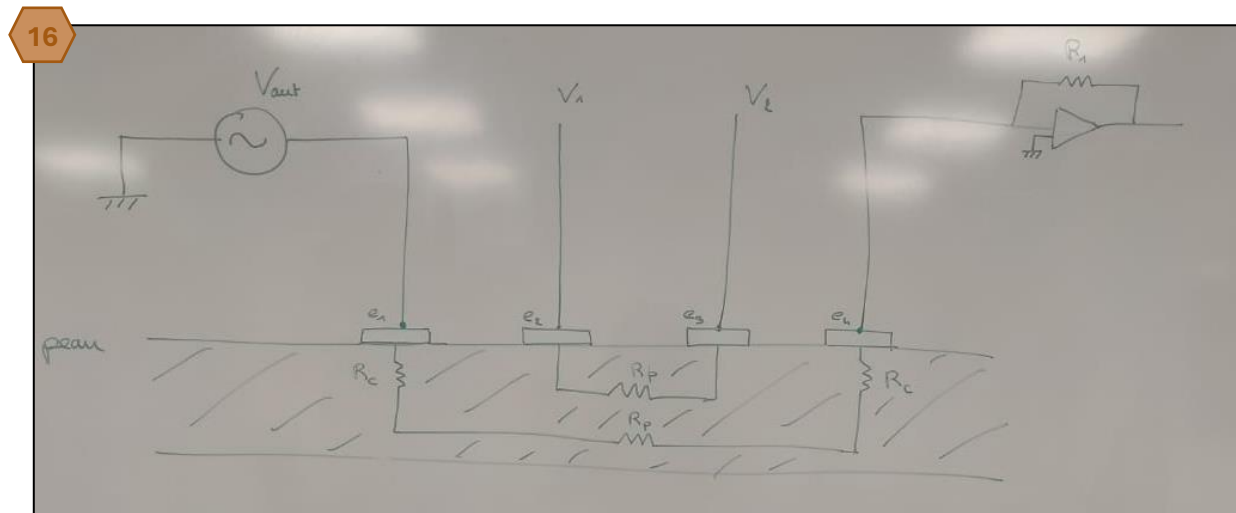
4 electrodes circuit V1
One "OUT" and one "IN"



4 electrodes circuit V2
One "OUT" and two "IN"

These are the theoretical circuits that we imagined with 4 electrodes. We tried 2 different ways. The first one uses only 1 "IN" and 1 "OUT". We now have 2 electrodes of alimentation and 2 electrodes of measurement. For the 2nd way, we used 2 "IN" and 1 "OUT".

But we realised later that those circuits were not correct for our measurements.



Final circuit for 4 electrodes

Now, "e1" and "e4" are wired the same as the circuit with 2 electrodes. They are called "extern electrode".

We added 2 other electrodes "e2" and "e3". These are called "intern electrodes". They are wired to the "IN" and "DIF" of the HF2LI Lock-In Amplifier. They are used to measure the "RESISTANCE OF CONTACT".

b) Theoretical calculus & results with 4 electrodes

The impedance of 2 electrodes is measured thanks to this formula:

For Z2e:

- $U = R.I$
- $I = U/R1$
- $I = \frac{U}{100}$

$$Z_{2e} = \frac{U}{I} = \frac{V_{out}}{I} = \frac{1}{I}$$

The impedance of 4 electrodes is measured thanks to this formula:

For Z4e:

- $U = R.I$
- $I = U/R1$
- $I = \frac{U}{100}$

$$Z_{4e} = \frac{U}{I} = \frac{V2 - V1}{I}$$

In our python code, we used some formulas to calculate the impedances and the phases.

17 `data["Z_2E_XY"] = Tension / ((data["Demod1 X (V)"] + 1j * data["Demod1 Y (V)"]) / Resistance) # Impédance complexe`

This is the formula we used to calculate the **complex impedance**:

$$I = \frac{(X + iY)}{R}$$

$$Imp.c = \frac{U}{I}$$

$$Imp.c = \frac{U}{\frac{X + iY}{R}}$$

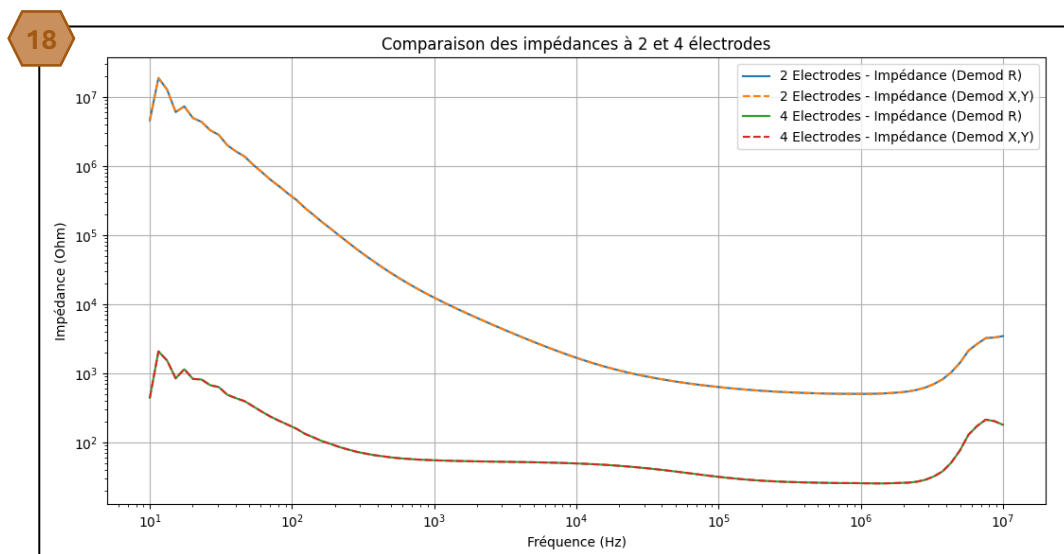
$$Imp.c = \frac{U \times R}{X + iY}$$

$$Imp.c = \frac{U \times R (X - iY)}{X^2 + Y^2}$$

$$Imp.c = \frac{U \times R \times X - (iU \times R \times Y)}{X^2 + Y^2}$$

$$Imp.c = \frac{U \times R \times X}{X^2 + Y^2} - i \left(\frac{U \times R \times Y}{X^2 + Y^2} \right)$$

This is what our python codes plots for the **impedance** with 4 electrodes:



Impedance measured and calculated for 1 person (with 2 & 4 electrodes)

We can see here that the impedance calculated with our python code (*red curve*) is the same as the impedance measured with LabONE (*green curve*). We can deduce that our calculus is correct.

To calculate the **phase**, we used the "argument" of the complex impedance:

$$\arctan\left(\frac{I}{R}\right)$$

19

```
data["Z_4E_Phase"] = data["Demod4 Phase (deg)"] - data["Demod1 Phase (deg)"]
data["Z_4E_Phase"] = np.unwrap(np.radians(data["Z_4E_Phase"])) * 180 / np.pi
data["Z_4E_XY_Phase"] = np.angle(data["Z_4E_XY"], deg=True)
data["Z_4E_XY_Phase"] = np.unwrap(np.radians(data["Z_4E_XY_Phase"])) * 180 / np.pi
```

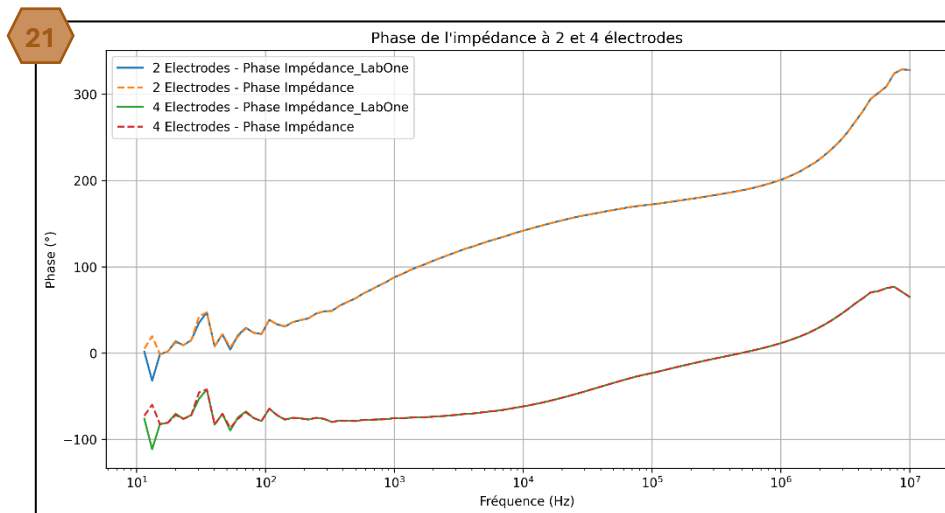
To calculate the **real impedance**, we used the argument of the complex impedance. This is the formula we used to calculate the real impedance:

$$Imp.r = \sqrt{\left(\frac{U \times R \times X}{X^2 + Y^2}\right)^2 + \left(\frac{U \times R \times Y}{X^2 + Y^2}\right)^2}$$

20

```
data["Z_2E_XY_Mag"] = np.abs(data["Z_2E_XY"]) # Magnitude de l'impédance
```

This is what our python codes plots for the **phase** with 4 electrodes:



Phase measured and calculated for 1 person (with 2 & 4 electrodes)

We can see here that the phase calculated with our python code (*red curve*) is the same as the phase measured with LabONE (*green curve*). We can deduce that our calculus is correct.

This is how our excel file looks like after all those calculi:

22

Whole excel sheet for one person

23

Close up on the same excel sheet

Frequency (Hz)	Demod R (V)	Demod Phase (deg)	Demod X (V)	Demod Y (V)	Demod R (V)	Demod Phase (deg)
10	0.00011836	-70.46451843	1.446035E-05	-0.00029896	0.00016687	-68.39382015
11.49756995	9.34796E-06	-45.14798906	6.40129E-06	-5.98952E-06	0.000131709	-3.67379007
13.21941148	1.79535E-05	-58.49024844	9.03888E-06	-1.42368E-05	0.000134122	-12.20649398
15.19511083	1.54444E-05	-54.89759586	7.99169E-06	-1.24315E-05	0.000136237	-19.2465881
17.475284	1.68763E-05	-62.19964649	7.76414E-06	-1.4857E-05	0.000130222	-27.52994772
20.09233003	1.67953E-05	-68.10075125	6.04306E-06	-1.53995E-05	0.000133905	-36.6377061
23.101297	3.33566E-05	-34.31655123	2.73821E-05	-1.87914E-05	0.000124099	-46.74424427
26.56087783	2.6791E-05	-45.78539009	1.7681E-05	-1.96113E-05	0.000113286	-59.1411761
30.53855009	1.1848E-05	-69.45791731	1.13489E-05	-2.86743E-05	0.00011346	-47.04708408
35.11191734	5.79563E-05	-59.29671833	2.95723E-05	-4.97435E-05	0.00010399	-77.33346011
40.57017259	5.7401E-05	-39.30465871	4.42201E-05	-5.82402E-05	9.95505E-05	-96.98070994
46.41588844	7.1551E-05	-39.5253004	5.48020E-05	-4.55587E-05	9.05908E-05	-106.524401
53.36699231	7.85961E-05	-50.65517021	4.95349E-05	-6.0691E-05	9.17296E-05	-121.6113142
61.35907273	0.000203844	-49.9455996	6.6703E-05	-7.95775E-05	0.000201589	-142.4708446
70.5480311	0.00013917	-39.06021291	0.000106387	-8.86756E-05	0.000110181	-152.2867516
81.11308308	0.000172462	-40.97384737	0.000129527	-0.000111236	0.000120121	-166.7493234
93.26033469	0.00025706	-45.96652288	0.000181254	-0.000181745	0.000147145	-177.3625509
107.2267222	0.000286759	-43.01607354	0.000209346	-0.000195255	0.00017327	-188.5856806
123.2846739	0.000391863	-46.00820492	0.00027193	-0.00028039	0.000257609	-197.8425174
141.7474163	0.000477373	-44.25981062	0.00031211	-0.000304325	0.000302646	-203.8488916
162.8750835	0.000562937	-45.526773	0.000392405	-0.000398887	0.000384612	-209.6597505

c) Resistance of contact with 4 electrodes

To find the resistance of contact in a circuit of 4 electrodes, this is the formula to use:

- $Z2e = 2Rc + Rp$
- $Z4e = Rp$
- ➔ $Rc = (Z2e - Z4e)/2$

VII. Results with 4 electrodes

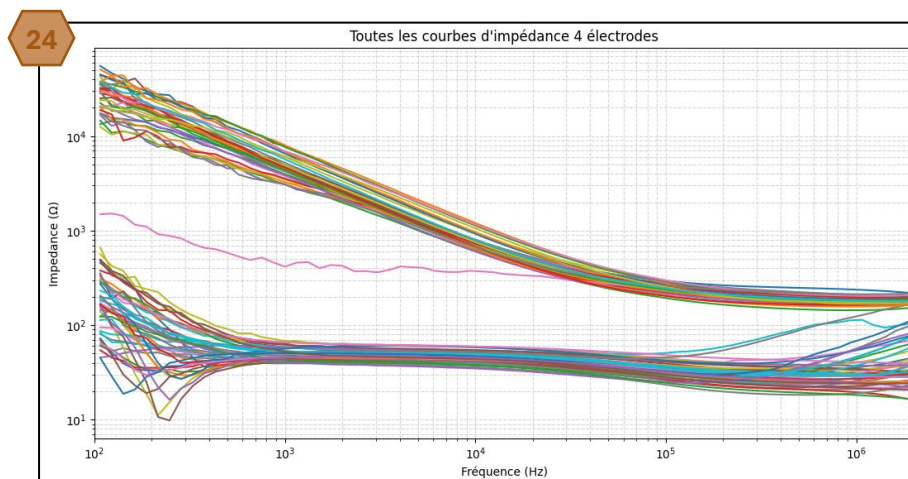
When we did some measurements on volunteers, we did 4 measurements:

- “**repos1**”: they had to keep their arm still
- “**contraction**”: they had to contract they're forearm
- “**mvt poignet**”: they had to move their wrist to create tension in their forearm
- “**repos2**”: they had to stay still once again, but this time, after having done a physical effort

We decided to take 4 different measurements per person for 2 reasons. First: build a larger database. Second: diversify the measurements, not only when volunteers are still but in movement.

a) “repos1”

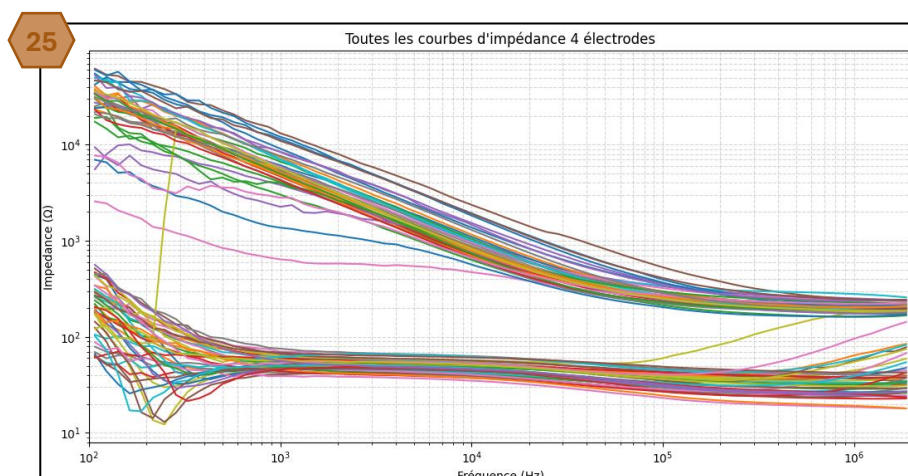
Concerning the impedance, we can observe that there are 2 main groups following 2 different tendencies. The first group is between **100 000 Ohms & 500 000 Ohms** and the second group is between **50 Ohms & 600 Ohms**.



Impedance curves for 81 people at the state “repos1”

b) “contraction”

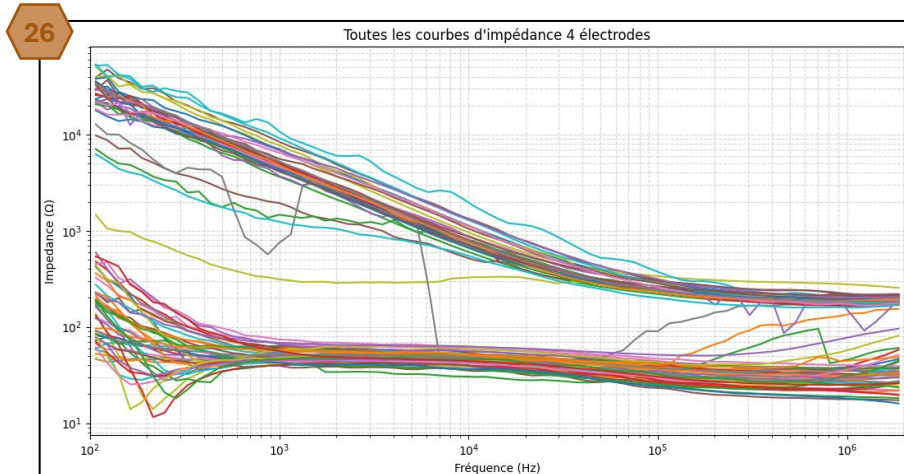
Concerning the impedance, we can observe that there are 2 main groups following 2 different tendencies. The first group is between **40 000 Ohms & 500 000 Ohms** and the second group is between **50 Ohms & 500 Ohms**.



Impedance curves for 81 people at the state “contraction”

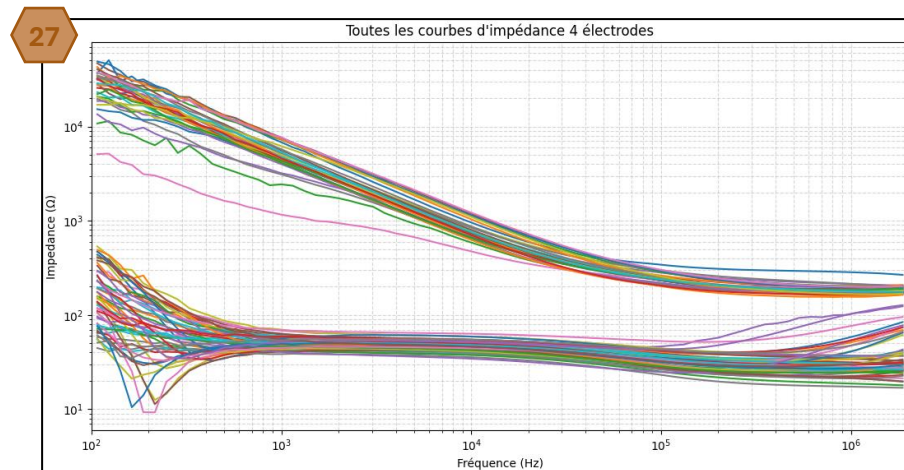
c) “mvt poignet”

Concerning the impedance, we can observe that there are 2 main groups following 2 different tendencies. The first group is between **50 000 Ohms & 400 000 Ohms** and the second group is between **50 Ohms & 500 Ohms**.



d) “repos2”

Concerning the impedance, we can observe that there are 2 main groups following 2 different tendencies. The first group is between **100 000 Ohms & 400 000 Ohms** and the second group is between **40 Ohms & 400 Ohms**.

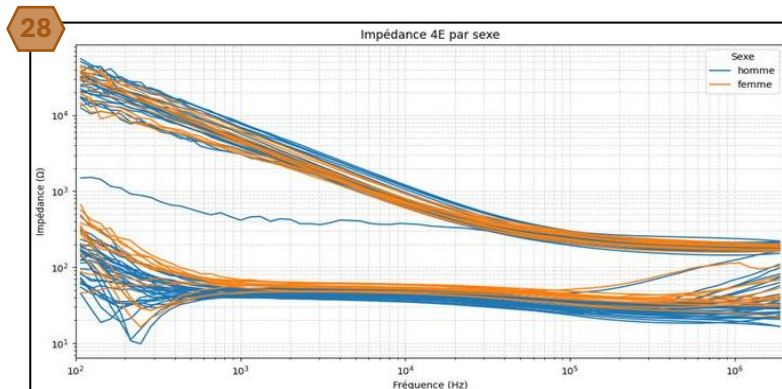


VIII. Interpretation of the 4 electrodes' results

As we noticed on the previous graphs, for each measurements there are two groups of curves. We wanted to know why there were two groups and how. We supposed that it was maybe due to 3 different characteristics. The first one being the sexe of the volunteer, the second one being the age and the last one being the monk scale.

a) With the "sexe"

First we tried by supposing that the two groups were because of the sexe of the volunteer. We represented the men with the blue curves. The women are represented with the red curve.



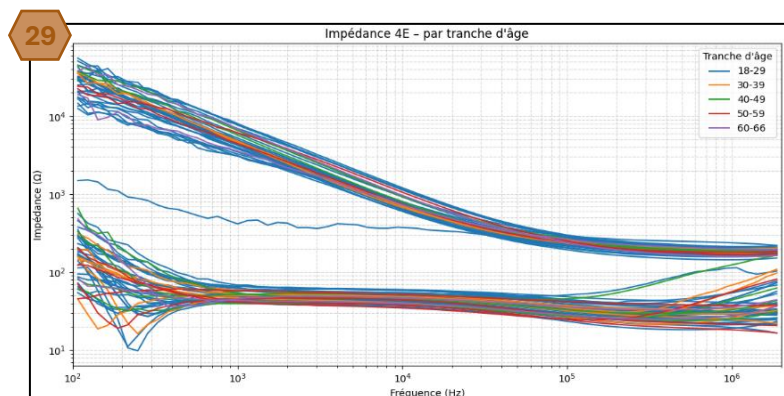
Impedance curves comparing "sexe"

But as we can see, there is not a group full of women and the other one full of men as we thought. The men and the women are mixed in both groups. So we can conclude that the sexe is not the parameter that caused these 2 different groups.

But, if we look closely, we can see that in the group with lower impedance, the women have a slightly a higher impedance than men. So we can conclude that if the client is a woman, she will have an higher impedance than a men client.

b) With the "age"

Secondly, we tried by supposing that the two groups were because of the age of the volunteers. We created 5 families of ages to have a more global view.

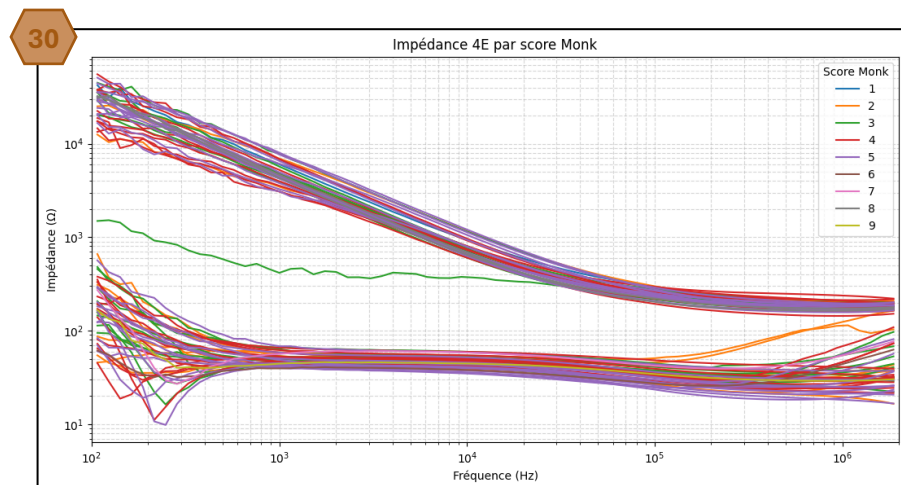


Impedance curves comparing "age"

Once again, we can see that all the different age groups are in both groups of curves. So those 2 groups are not linked to the volunteers' age.

c) *With the “monk scale”*

Finally, we tried by supposing that the two groups were because of the color of the skin of the volunteer.



Once again, we can see that all the different monk scales are in both groups of curves. So those 2 groups are not linked to the volunteers' skin color.

Unfortunately, we do not really know how to interpret ethos two groups of impedance. Maybe it is due to other biometrics that we have not tested. Or maybe there is the difference between people who have been measured before lunch and others who have been measured after lunch.

IX. AI

a) Database

The first step before implementing our AI was to build a solid database with all the data we measured and collected from our 83 volunteers.

The two types of data we had to take care of are:

- Data measured on their skin (phase, impedance) and 2 biometrics (weight, height)
- Biometrics from them (age, sexe, monk scale)

IMPORTANT: those two “families” will be our 2 entries of our database:

- *x entry*: phase, impedance, weight, height
- *y entry*: age, sexe, monk scale

Each measurement has been stocked in a csv file. So, we had 324 files to deal with.

When we made volunteers fill in the data module, we entered their information in an excel file by also writing the name of each csv file corresponding to each person.

Then, to link everything together, we used a Python code thanks to which we obtained our final database.

Each line corresponds to a volunteer. Each column corresponds to a biometric or a data value belonging to that same volunteer.

b) Finding the best model

Before choosing our final AI, we tested on different models to see which one was the best with the type of data that we had. We tested 5 different models with our data measured on volunteers (81 lines). The models are:

- Logistic Regression model
- Random Forest model
- Gaussian model
- K-Neighbors Classifier model
- SVC model

First, we tested with the “**LogisticRegression**” model.

```
X_train, X_test, y_train, y_test = train_test_split(
    X, y,
    test_size=0.2,
    random_state=42,
    stratify=y
)
print(f"\nTaille train: {len(X_train)}, test: {len(X_test)}")

# Choix du modèle (exemple : Logistic Regression)
chosen_model = Pipeline([('scaler', scaler), ('lr', LogisticRegression(max_iter=1000))])
chosen_model.fit(X_train, y_train)

y_pred = chosen_model.predict(X_test)
print("\n== Évaluation sur le set de test ==")
print("Accuracy:", accuracy_score(y_test, y_pred))
print("\nClassification Report:\n", classification_report(y_test, y_pred, target_names=le.classes_))
print("\nConfusion Matrix:\n", confusion_matrix(y_test, y_pred))
```

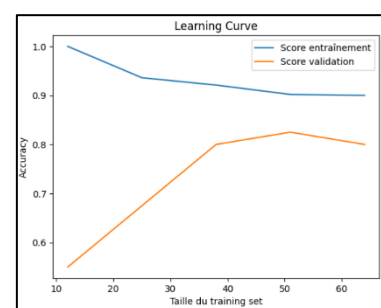
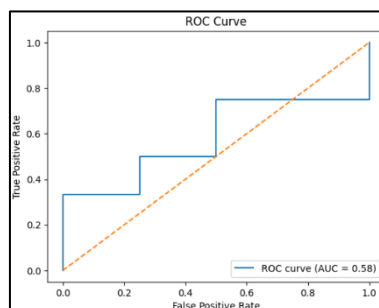
```
Taille train: 64, test: 16
== Évaluation sur le set de test ==
Accuracy: 0.625

Classification Report:
      precision    recall  f1-score   support

   Femme      0.33      0.50      0.40         4
   Homme      0.80      0.67      0.73        12

 accuracy      0.62         16
  macro avg     0.57      0.58      0.56         16
  weighted avg     0.68      0.62      0.65         16

Confusion Matrix:
[[2 2]
 [4 8]]
```



We had an accuracy of **60%** but as we have a lot less data for women, the model struggles to detect when the patient is a woman.

Then, we tested the “**RandomForest**” model.

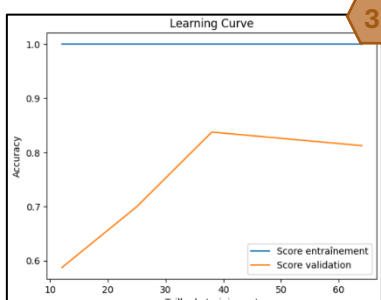
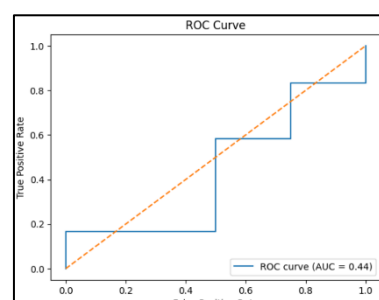
```
Taille train: 64, test: 16
== Évaluation sur le set de test ==
Accuracy: 0.5

Classification Report:
      precision    recall  f1-score   support

   Femme      0.17      0.25      0.20         4
   Homme      0.78      0.58      0.64        12

 accuracy      0.50         16
  macro avg     0.43      0.42      0.42         16
  weighted avg     0.57      0.50      0.53         16

Confusion Matrix:
[[1 3]
 [5 7]]
```



We obtained an accuracy of **50%**, which is not good. We can also see in the metrics that the model is better trained on men data compared to women data (precision: 0.17 / recall: 0.25 / f1_score: 0.20).

Then we tested with the “**Gaussian**” model.

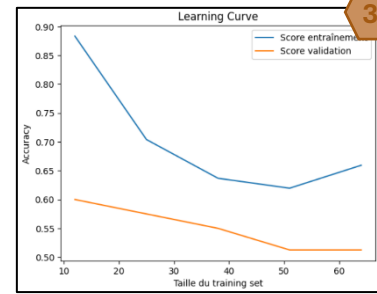
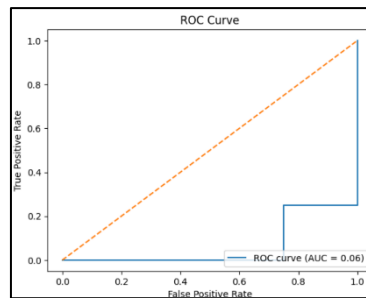
```
Taille train: 64, test: 16
== Évaluation sur le set de test ==
Accuracy: 0.25

Classification Report:
      precision    recall  f1-score   support

   Femme      0.00      0.00      0.00         4
   Homme      0.50      0.33      0.40        12

 accuracy      0.25
 macro avg      0.25      0.17      0.20        16
weighted avg      0.38      0.25      0.30        16

Confusion Matrix:
[[0 4]
 [8 4]]
```



33

The accuracy is at **25%**, catastrophic. With the metric, we see clearly that the trained data for women are not even taken in account (precision: 0.00 / recall: 0.00 / f1_score: 0.00).

Then, we tested the “**KNeighborsClassifier**” model.

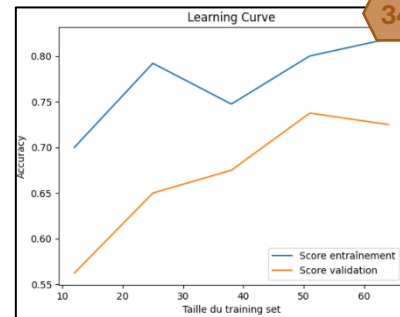
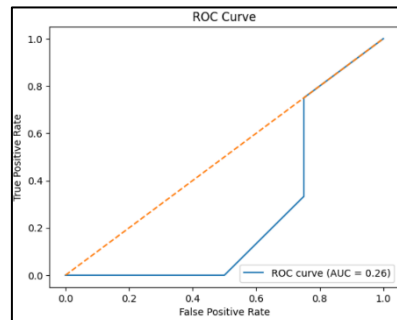
```
Taille train: 64, test: 16
== Évaluation sur le set de test ==
Accuracy: 0.625

Classification Report:
      precision    recall  f1-score   support

   Femme      0.25      0.25      0.25         4
   Homme      0.75      0.75      0.75        12

 accuracy      0.62
 macro avg      0.50      0.50      0.50        16
weighted avg      0.62      0.62      0.62        16

Confusion Matrix:
[[1 3]
 [3 9]]
```



34

The accuracy was at **62%**. But the results were not satisfying. Here, the women data are once again less well trained than men data (precision: 0.25 / recall: 0.25 / f1_score: 0.25)

Then we tested the “**SVC**” model.

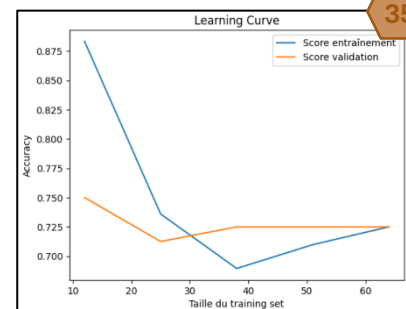
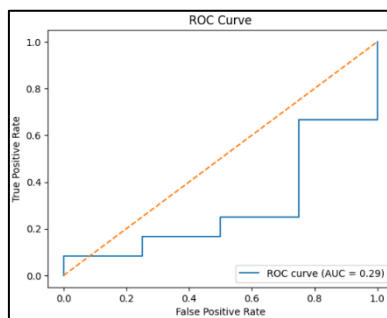
```
Taille train: 64, test: 16
== Évaluation sur le set de test ==
Accuracy: 0.5625

Classification Report:
      precision    recall  f1-score   support

   Femme      0.00      0.00      0.00         4
   Homme      0.69      0.75      0.72        12

 accuracy      0.56
 macro avg      0.35      0.38      0.36        16
weighted avg      0.52      0.56      0.54        16

Confusion Matrix:
[[0 4]
 [3 9]]
```



35

The accuracy was at **56%**. And the prediction seemed to be good. But the women data were not even taken in account (precision: 0.00 / recall: 0.00 / f1_score: 0.00).

As we measured data ourselves with 83 volunteers, and 4 measurements for each person, we don't have a large database. This is why we decided to randomize our data. We took all our data and applied a randomizer of $[-0.1, +0.1]$. So we went from 80 lines to 160 lines.

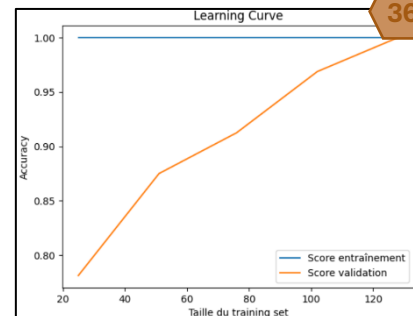
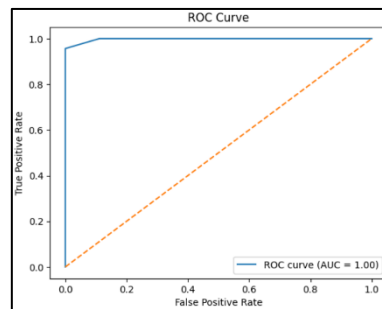
```
Taille train: 128, test: 32
== Évaluation sur le set de test ==
Accuracy: 0.9375

Classification Report:
              precision    recall  f1-score   support

   Femme      0.82      1.00      0.90         9
   Homme      1.00      0.91      0.95        23

 accuracy      0.91      0.96      0.94        32
 macro avg      0.91      0.96      0.93        32
 weighted avg    0.95      0.94      0.94        32

Confusion Matrix:
[[ 9  0]
 [ 2 21]]
```



```
Colonnes du nouveau fichier : ['Age', 'Taille', 'Poids', 'Impedance_107_repos1', 'Impedance_123_repos1', 'Impedance_141_repos1', 'Impedance_162_repos1']
Sexe_prédit
0      Homme
1      Homme
2      Femme
```

Then, we decided to randomize twice the data for the men and 4 times the data for women in the aim to have approximatively 50% men data and 50% women data (which was not the case until now because we had 70% men data and 30% women data).

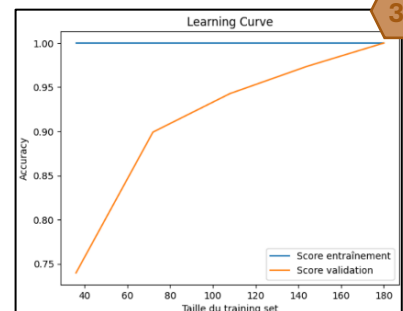
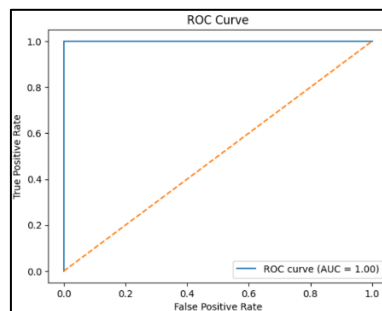
```
Taille train: 180, test: 46
== Évaluation sur le set de test ==
Accuracy: 1.0

Classification Report:
              precision    recall  f1-score   support

   Femme      1.00      1.00      1.00        22
   Homme      1.00      1.00      1.00        24

 accuracy      1.00      1.00      1.00        46
 macro avg      1.00      1.00      1.00        46
 weighted avg    1.00      1.00      1.00        46

Confusion Matrix:
[[22  0]
 [ 0 24]]
```



```
Colonnes du nouveau fichier : ['Age', 'Taille', 'Poids', 'Impedance_107_repos1', 'Impedance_123_repos1', 'Impedance_141_repos1', 'Impedance_162_repos1']
Sexe_prédit
0      Homme
1      Homme
2      Femme
```

The problem we encountered here is **overfitting**. The model has been trained on values that are too similar one to another, that is why we obtain perfect accuracy (**100%**).

c) Increasing the database

So we decided to use another way of creating new values to increase our database. For that, we used the “**SMOTE**” method. This model takes the samples we already have and it looks at the entire distribution of the data we have. Then, he creates samples of the data we have (age, weight, height, impedances), and for that, he relies on the samples already created. Essentially, in the lines of the Excel sheet, he will base himself, for example, between two neighboring lines and create a new line that connects with the line below and above.

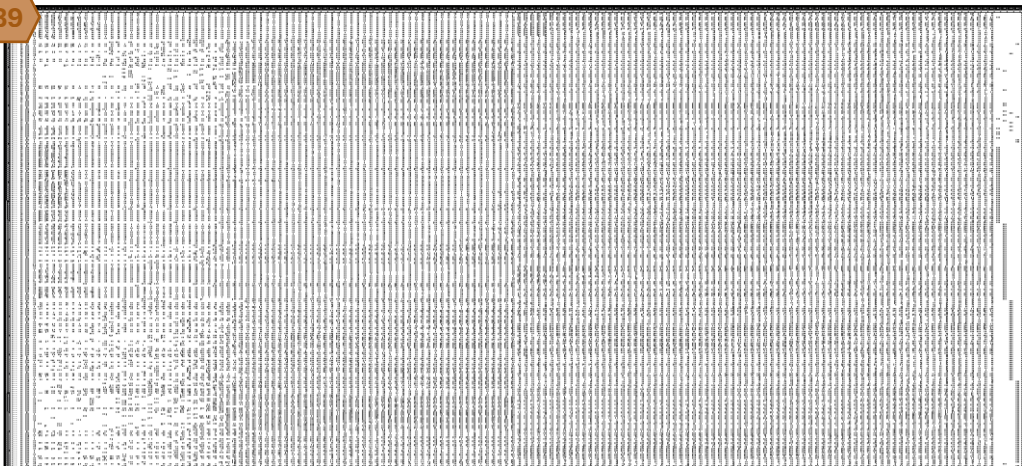
This method allowed us to have a much huger database (from 81 lines to 289 lines). Thanks to this, we will be able to train our AI even better. On the following pictures, we cannot see clearly the data but it is just to understand how much we increased it.

38



Original database with only measured data

39



Final database with new values created with SMOTE

d) Implementing the AI (V1)

We decided to test different AI models for each biometric to be sure we had great accuracy (so a good detection) for each biometrics independently.

First, we started with the “sexe” because it is the easier one because it is a binary result (WOMAN or MAN).

```
--- Début de l'évaluation cross-val pour RandomForest (Sexe) ---
évaluation_cross_val_pour RandomForest (Sexe)
accuracy CV : 0.792 (+/- 0.103)
f1_weighted CV : 0.783 (+/- 0.111)
precision_weighted CV : 0.818 (+/- 0.082)
recall_weighted CV : 0.792 (+/- 0.103)

--- Division des données pour Sexe ---
Taille train : 230, test: 58
Distribution des classes dans l'entraînement pour Sexe:
0 115
1 115
Name: count, dtype: int64
Distribution des classes dans le test pour Sexe:
0 29
1 29
Name: count, dtype: int64
```

== Résultats de l'évaluation de classification pour Sexe sur le set de test == 40

Accuracy: 0.79

	precision	recall	f1-score	support
Femme	0.77	0.83	0.80	29
Homme	0.81	0.76	0.79	29
accuracy			0.79	58
macro avg	0.79	0.79	0.79	58
weighted avg	0.79	0.79	0.79	58

Confusion Matrix:

```
[[24  5]
 [ 7 22]]
```

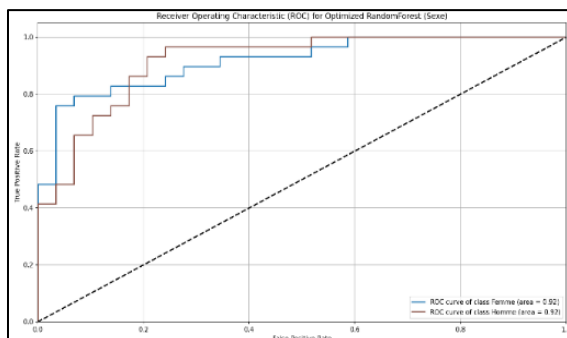
For the “sexe”, we obtained an **accuracy of 79%**, which is good. Also, if we look at the results for each sexe, we see that the results are not far from each other and always **>75%**, meaning that the model learned equally for men data and for women data:

Probabilités de prédiction pour chaque classe (vérifier l'ordre de l'encoder si besoin):

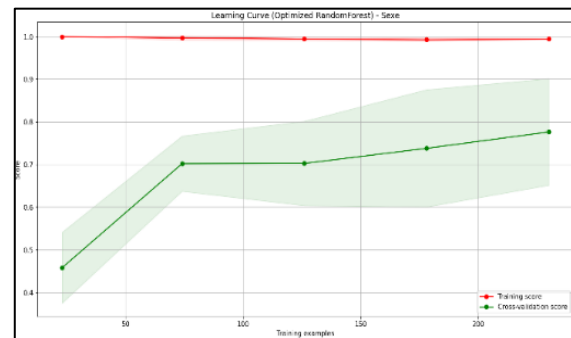
Femme: 0.5841

Homme: 0.4159

- Training score: converging to **100%**
- Cross-validation score: converging to **almost 80%**



ROC curve for the “sexe”



Learning curve for the “sexe”

Once our results to predict the “sexe” of a client were satisfying, we focused on predicting the age of a client. But as this biometric is nonbinary and predicting the exact age seemed to be a little ambitious, we decided to cut the ages in different age groups:

- [18;24]
- [25;31]
- [32;38]
- [39;45]
- [46;52]
- [53;59]
- [60;66]

--- Début de l'évaluation cross-val pour RandomForestClassifier (Tranche_Age) -

41

Évaluation cross-val pour : RandomForestClassifier (Tranche_Age)

accuracy CV : 0.771 (+/- 0.071)

f1_weighted CV : 0.768 (+/- 0.074)

precision_weighted CV : 0.787 (+/- 0.071)

recall_weighted CV : 0.771 (+/- 0.071)

Cross-validation: 5 trainings, for each training the data will be divided in 80%-20% differently

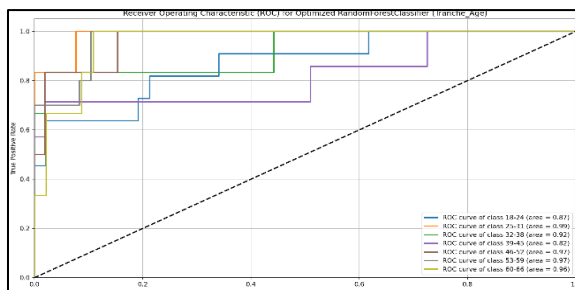
```
--- Évaluation sur le set de test (Tranche_Age) ---
```

== Résultats de l'évaluation de classification pour Tranche_Age sur le set de test ==

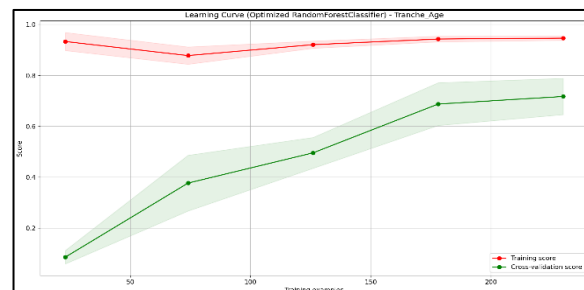
Accuracy: 0.76

Classification Report:				
	precision	recall	f1-score	support
18-24	0.64	0.64	0.64	11
25-31	0.83	0.83	0.83	6
32-38	0.83	0.83	0.83	6
39-45	0.83	0.71	0.77	7
46-52	0.71	0.83	0.77	6
53-59	0.78	0.70	0.74	10
60-66	0.77	0.83	0.80	12
accuracy			0.76	58
macro avg	0.77	0.77	0.77	58
weighted avg	0.76	0.76	0.76	58

Here, we also obtain great accuracy of **76%**, which means that our model predicts well the age group of the client tested.



ROC curve for the “age”



Learning curve for the “age”

By looking at the curves, we can once again see that our model is well trained and is not overfitting. For example, if we have a look at the learning curve:

- Training score: converging to **95%**
- Cros-validation score: converging to **70%**

The last biometric we had to deal with was “the Monk scale of the skin”.

For this, we have 10 groups:

- 1	Monk	Monk	Monk	Monk	Monk	Monk	Monk	Monk	Monk
- 2	01	02	03	04	05	06	07	08	09
- 3									
- 4									
- 5									
- 6									
- 7									
- 8									
- 9									
- 10									
	#f6ede4	#f3e7db	#f7ead0	#eadaba	#d7bd96	#ao7e56	#825c43	#604134	#3a312a
									#292420

42 == Résultats de l'évaluation de classification pour la TEINTE DE PEAU ==

Accuracy (Monk): 0.71

Classification Report (Monk):

	precision	recall	f1-score	support
1	0.00	0.00	0.00	0
2	0.65	0.79	0.71	14
3	0.82	0.67	0.74	21
4	0.68	0.76	0.72	17
5	1.00	0.50	0.67	6
8	0.00	0.00	0.00	0
accuracy			0.71	58
macro avg	0.53	0.45	0.47	58
weighted avg	0.76	0.71	0.72	58

Here, we also obtain great accuracy of **71%**, which let us think that our model predicts well the MONK scale of the client tested. But, as we can see, only 6 classes out of 10 are considered by AI. Also 2 of the scales predicted have a precision of **0%**. This is because in our database we do not have enough data for the monk scale 1, 6, 7, 8, 9, 10.

== Prédiction pour les échantillons sélectionnés de 'Data_1.xlsx' ==			
	Sexe_prédit	Tranche_Age_prédite	Monk_prédit
0	Homme	53-59	5
1	Homme	25-31	2
2	Femme	18-24	4

Here, are some predictions that we obtained with our 3 models of detection together (sexe, age, monk). We obtain a perfect accuracy for the sexe and the age of the patient. However, all the predictions for the monk scale are false:

- Monk for 0: predicted=5 | real=4
- Monk for 1: predicted=2 | real=6
- Monk for 2: predicted=4 | real=2

This could be explained by the difficulty of having an appropriate model for the three biometrics at the same time. Indeed, this was quite hard to configure a precise model for the sex, the age and monk in terms of for example, numbers of trees or depth of the trees...

e) Implementing the AI (V2)

As we saw that the results were not convincing, we decided to change our way of increasing the amount of data in our database. For that, we choose to randomize our values.

First, we doubled the number of men to have more people. And we did a times 4 for the women (as we had fewer data from them). At the same time, we made sure to keep the biometrics intact (sexe, age, monk scale). We applied a delta of +/- 5% on the impedance, phase, weight and height values.

As we saw that the results were not convincing, we decided to change our way of increasing the amount of data in our database. For that, we choose to randomize our values.

First, we doubled the number of men to have more people. And we did a times 4 for the women (as we had fewer data from them). At the same time, we made sure to keep the biometrics intact (sexe, age, monk scale). We applied a delta of +/- 5% on the impedance, phase, weight and height values.

After building a new database, we also decided to change our AI model (from “random forest” to a “neural network”). We chose a neural network because it is more precise even though it is not stable (the opposite of the random forest model). Also, it allows a better generalization than the random forest model.

For this trial, we thought that the best way to have a good model was to have 3 different models for each biometrics we want to detect. The model is the same: **multi-layer perceptron**. We implemented it 3 times because a single model cannot output the 3 data at the same time.

First, we tried to test our new model just by predicting the sexe and the age of a patient. We got satisfying results:

- **90%** of accuracy for the **sexe** on our data sample
- **50%** of accuracy for the **age** on our data sample

Then, we tried to test our new model by predicting the sexe, the age and the monk scale of a patient. We got fewer satisfying results:

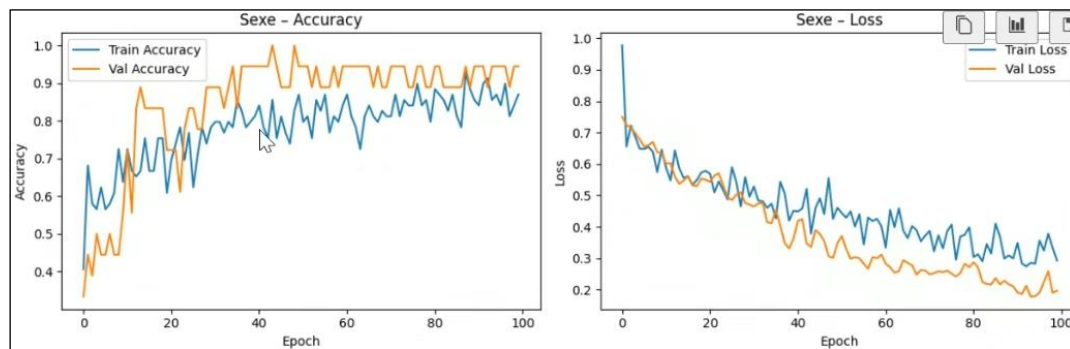
- **90%** of accuracy for the **sexe** on our data sample
- **20%** of accuracy for the **age** on our data sample
- **20%** of accuracy for the **monk** on our data sample

Sexe_Predit	AgeGroup_Prec	MonkGroup_Pr	Sexe_Predit	AgeGroup_Prec	MonkGroup_Pr
Femme	18-29	3-4	Femme	50-59	3-4
Homme	18-29	5-6	Homme	18-29	5-6
Femme	18-29	3-4	Femme	50-59	3-4
Homme	18-29	5-6	Homme	30-39	5-6
Homme	18-29	1-2	Homme	60-69	1-2
Femme	50-59	3-4	Femme	30-39	3-4
Homme	50-59	5-6	Homme	40-49	5-6
Femme	18-29	5-6	Femme	20-29	5-6
Homme	50-59	5-6	Homme	40-49	5-6
Femme	50-59	5-6	Femme	50-59	5-6

Predictions for the sexe, the age, the monk scale with our final model

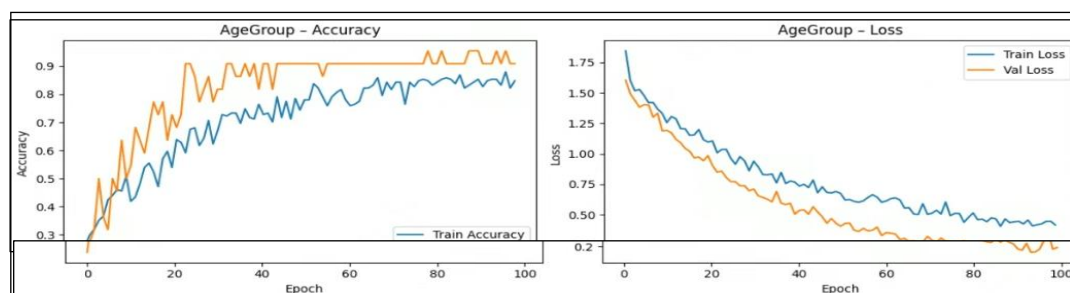
Once again, we encountered difficulties with the prediction of the monk scale, which impacted badly the prediction of the other biometrics (especially the age).

Concerning the **sexe**, we have an accuracy of **~90%** with a loss of **~30%**. Which is good and corresponding to the prediction the model made. Also, the curves representing the “train accuracy” and the “validation accuracy” are following the same tendency, which is good.



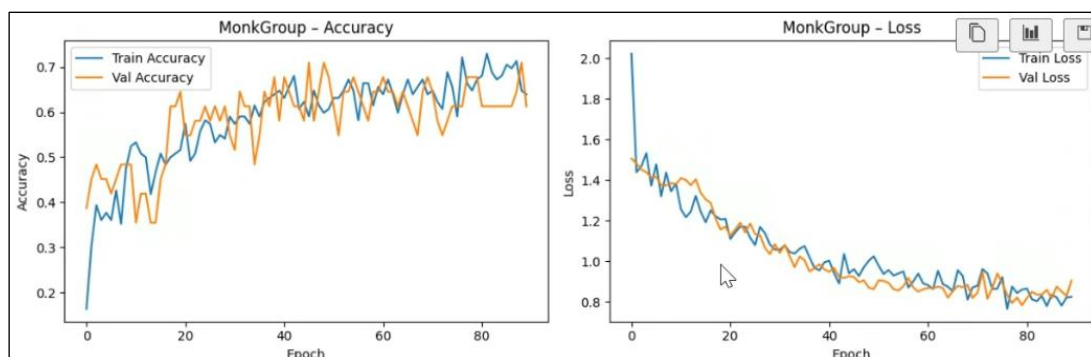
Accuracy and loss curves for the “sexe”

Concerning the **age**, we have an accuracy of **~80%** with a loss of **~20%**. Which is good but not corresponding to the prediction the model made. Also, the curves representing the “train accuracy” and the “validation accuracy” are following the same tendency, which is good



Accuracy and loss curves for the “age”

Concerning the **monk scale**, we have an accuracy of **~60%** with a loss of **~80%**. Which is terrible and corresponding to the prediction the model made. However, the curves representing the “train accuracy” and the “validation accuracy” are following the same tendency, but the prediction are not good at all, and the loss rate is high. The model is not overfitting.



Accuracy and loss curves for the “monk scale”

f) Implementing the AI (Vfinal)

To be fair, we tested this model before the one we just presented. But as we saw that this one had a better prediction compared to the one we just presented, we decided that it will be our final version of our AI.

For the database, we stayed as the previous model with a database augmented with the randomization method.

First, we tried to test our new model just by predicting the sexe and the age of a patient. We got satisfying results:

- **90%** of accuracy for the **sexe** on our data sample
- **60%** of accuracy for the **age** on our data sample

Then, we tried to test our new model by predicting the sexe, the age and the monk scale of a patient. We got fewer satisfying results:

- **80%** of accuracy for the **sexe** on our data sample
- **50%** of accuracy for the **age** on our data sample
- **20%** of accuracy for the **monk** on our data sample

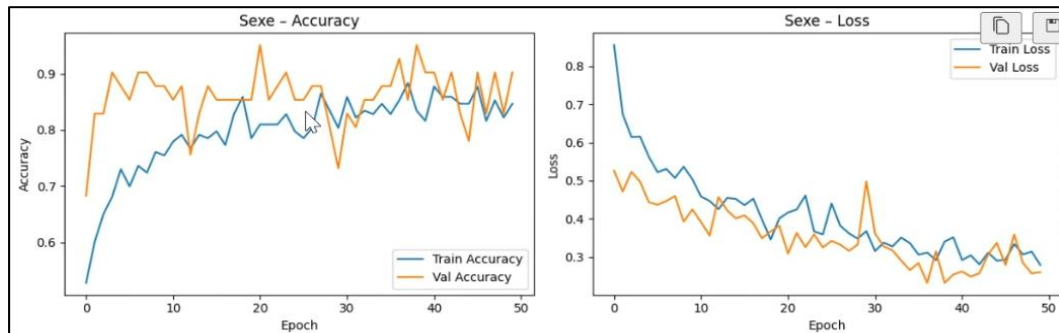
Sexe_Predit	AgeGroup_Pred	Sexe_Predit	AgeGroup_Prec	MonkGroup_Pr
Femme	18-29	Femme	18-29	5-4
Homme	18-29	Homme	18-29	5-6
Femme	18-29	Femme	18-29	5-4
Homme	30-39	Homme	18-29	5-6
Homme	18-29	Homme	18-29	5-6
Femme	18-29	Femme	18-29	5-4
Homme	50-59	Homme	18-29	5-6
Femme	18-29	Femme	18-29	5-6
Homme	50-59	Homme	40-49	5-6
Femme	30-39	Homme	18-29	5-6

Predictions for the sexe, the age, the monk scale with our final model

Once again, we encountered difficulties with the prediction of the monk scale, which impacted badly the prediction of the other biometrics.

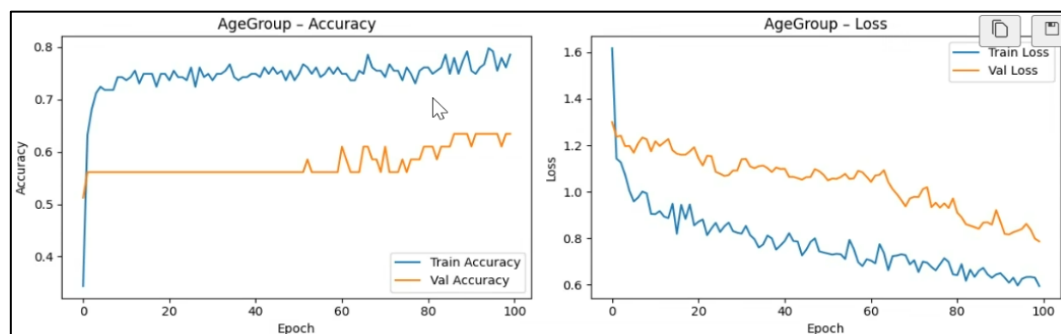
Here are the curve representing the accuracy and the loss for each biometric. In general, we can see that the results are not that bad. But if we have a closer look, we can see some malfunctions.

Concerning the **sexe**, we have an accuracy of **~85%** with a loss of **~30%**. Which is good and corresponding to the prediction the model made. Also, the curves representing the “train accuracy” and the “validation accuracy” are following the same tendency, which is good.



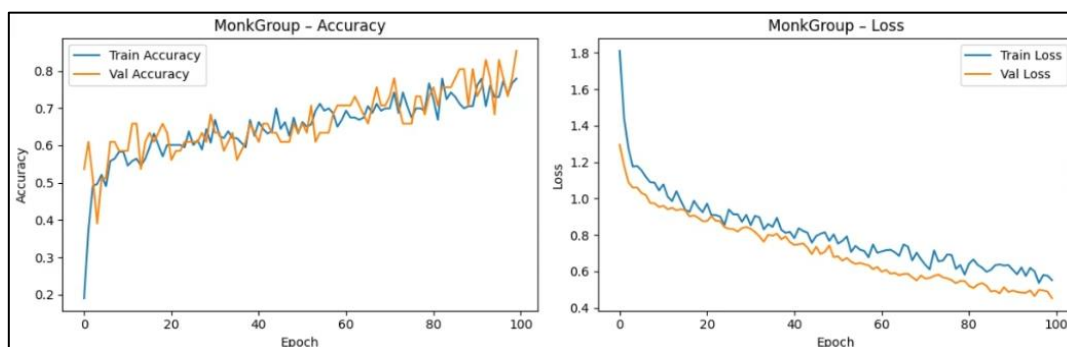
Accuracy and loss curves for the “sexe”

Concerning the **age**, we have an accuracy of **~60%** with a loss of **~80%**. Which is terrible and corresponding to the prediction the model made. Also, the curves representing the “train accuracy” and the “validation accuracy” are not following the same tendency, which represents an overfitting. But the predictions are bad, which seems weird.



Accuracy and loss curves for the “age”

Concerning the **monk scale**, we have an accuracy of **~80%** with a loss of **~50%**. Which is not that good and corresponding to the prediction the model made. However, the curves representing the “train accuracy” and the “validation accuracy” are following the same tendency, but the prediction are not good at all, and the loss rate is high. The model is not overfitting.



Accuracy and loss curves for the “monk scale”

X. Our problems and solutions

One of the major problems that we encountered at the beginning of the second semester was how to build our electronic circuit. One of our project sponsors Gaëlle MONTEIL helped us on this part so we can have a functional circuit.

Another issue is after doing several tests and measurements with 2 electrodes, our projects sponsors were not sure about using 4 electrodes or staying with 2 electrodes. This reflexional time took them a while so we could not really move forward with this project. We just tried to imagine how to transform our 2 electrodes circuit into a 4 electrodes circuit. We also took other measures to not waste much time.

Once we knew that we had to continue the experiments with a 4 electrodes circuit, we decided to take some measurement on volunteers. We prepared the consent paper and a google form to get the biometrics from the volunteers. The problem was that it was complicated to get the data from different ages. It was easier to get students than teachers or people from the administration. To diversify our data, we send mails to teachers and the administration, thanks to that, we were able to have a more diversified database.

Once we got all our volunteers tested with all their data, we could plot their impedance curve to have a better view. We observe that there were 2 different groups: one with higher impedance than the other. We wanted to know which parameter caused this, but we did not find why. Maybe we should have tested more parameters to get our answer, but we preferred go forward in the project.

One of the other problems was having a database compatible with an AI model. We used a Python code to build our final database. We took all the volunteers' data that was stored in csv file and put it in an excel file. Also, we took the excel file containing all the biometrics of the volunteers and only a few to put them in the same excel sheet as the previous csv files, and each biometrics has been linked to their measured values. Like that, we obtained a perfect database.

After that, we decided to make our database bigger as we only had 81 lines of data. We used a model called "randomize" to create new data from the data already existing. Thanks to that, we obtained a database of 289 lines, which let us have a better database to train our AI even better.

Our last problem was to implement a good AI, avoiding overfitting. We tested our training data with different models to pick the best one. We finally chose the random forest model with its accuracy of 90% for the sexe detection and 60% for the age detection.

XI. Conclusion

During this last semester, we focused on the IRL experiments part of our master project.

We first built a RC circuit to have a better understanding of how to build the circuit and get our first IRL impedance curve.

Then, we kept the same circuit but added 2 electrodes to take real measurements of the impedance directly on our skin. We also plotted the impedance curve to see with which label (from our theoretical experiments) our IRL measurements were corresponding.

After that, we added 2 other electrodes because it allowed us to have more precise data. We took several measures from 83 volunteers to build our database for our AI. This part took us a while to do because taking the measurements on each volunteer was taking 10-15minutes to do (the time to make them sign the consent paper, place the electrodes, take the data, save them in the csv file, take back their biometrics in an excel file).

The final step of this project was to implement our AI. Before choosing a model, we tested several methods to decide which one was the best with our database. Also, this database was not huge enough, we used a technique (randomization) to create new data based on real data that we had measured. We went from 81 line of data to 289, which is a great improvement. It allowed us to have better trained AI. We finally chose the random forest model, which gave us an AI of 90% accuracy for the sexe detection and 60% for the age detection.